Franklin Fuchs

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EDUCATION

Johns Hopkins University **2021 – 2023**
MS in Biostatistics

University of Nevada, Reno **2017 – 2021**
Honors BS in Mathematics (Statistics Emphasis), Minor in Big Data

SKILLS

- **Programming Languages:** Python, SQL, R, C++, C, MATLAB
- **Software:** GCP, AWS, Databricks, BigQuery, Dataiku, Tableau, Power BI, Snowflake, Docker, Singularity, SLURM, Markdown, Shiny, LaTeX, Linux Shell, Office Suite, Git, and GitHub
- **Modeling and Computing:** Multi-Objective Optimization, Classification, Regression, Time Series Forecasting, Predictive Analytics, Feature Engineering, Data Wrangling, Variable Selection, Exploratory Data Analysis, Data Web-App Development, Database Design, Interactive Data Visualization, LLM Infrastructure Development

WORK

Mondelez Insights and Analytics Group **Jun. 2025 – Present**
Business Intelligence and Forecasting Manager ← Data Science Manager

- Quantified the impact of GLP-1 medication adoption on our total (snacking) product portfolio by integrating prescription claims, Nielsen retail sales, and macroeconomic data across 9 U.S. Census Divisions; built cohort decay and Explainable Boosting Machine (EBM) models isolating GLP-1 as a statistically significant but secondary headwind directly informing category strategy.
- Built an automated customer-level sales forecasting system for 5 snacking categories (~$2B+ revenue) across 27 retail customers, migrating legacy Excel processes to Databricks, PySpark, and BigQuery. Designed anchored disaggregation methodology aligning forecasts with executive-approved channel targets. Established Git-based version control and modular Python packaging for reproducible analytics.
- Developed an Elastic Net model quantifying halo dollar impact of the LTE Oreo portfolio ($150M revenue) on total Oreo brand revenue ($2.4B) over 2.5 years. Recommended portfolio adjustments to maximize marginal revenue per marginal TDP change, using shrinkage methods for multicollinearity and variable selection across Economic, Panel, Price, Promotion, Distribution, and Category data.
- Built a synthetic matching framework with Streamlit interface to optimize Oreo in-store placement using KNN store matching and ANOVA testing. Integrated multi-source retail data for pricing, promotion, and product strategy through covariance analysis and overlap diagnostics.

Pfizer AI, Data, Analytics, Digital Health Architecture Groups **Jun. 2023 – Jun. 2025**
Data Scientist ← Software/Solution Engineer (Rotational Associate)

- Created static and time-dependent recommender/ranking systems for Target Product Profile Benefit-Risk Modeling using Multi-Objective Optimization (Evolutionary Algorithms, Reinforcement Learning, Robust Portfolio Optimization) in Python and R.
- Designed interactive visualizations highlighting unmet needs across demographic patient groups and diseases, using SQLite/MS SQL Server for storage, AWS for deployment, and custom HTML/CSS for UI styling.
- Enhanced internal Python data APIs with LLM/Chatbot capabilities for accelerated data ingestion, building RAG systems with source citation and tabular dataset generation using specialized LLM-based agents.
- Conducted Non-Parametric Statistical Modeling to support indication expansion of current portfolio drugs, integrating efficacy, safety, viability, and patient preference data.
- Prototyped Vector Database and RAG features for the internal LLM platform with enterprise scalability focus, using AWS cloud tools and delivering results as a Python/Streamlit web application.